

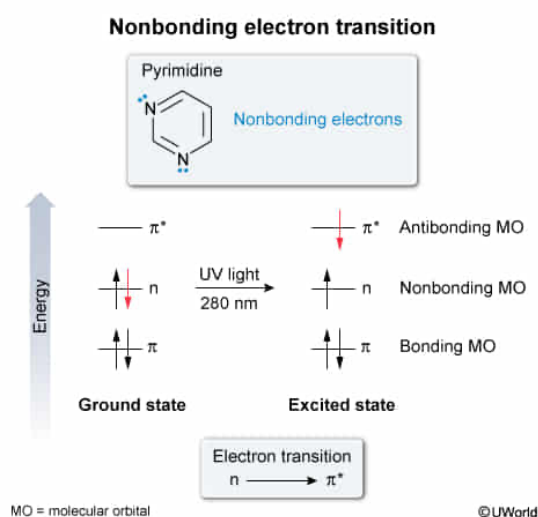
In the ultraviolet (UV) range, pyrimidine has maximum absorptions at 240 nm due to π electron excitation and at 280 nm due to nonbonding electron excitation. The nonbonding electrons on nitrogen in pyrimidine undergo which of the following transitions when absorbing UV light?

- ☐ A. $\pi \rightarrow \pi^*$ [35%]
☒ B. $n \rightarrow \pi^*$ [39%]
☐ C. $\pi^* \rightarrow \pi$ [14%]
☐ D. $\pi^* \rightarrow n$ [9%]

Correct

39% answered correctly

Explanation:



The **electromagnetic spectrum** consists of waves of varying wavelengths, energy, and frequency. **Ultraviolet (UV) light**, which corresponds to wavelengths **between 200 nm and 400 nm**, is used in UV spectroscopy.

In UV spectroscopy, a compound in solution is irradiated with UV light; the amount of UV light absorbed by the compound is measured at each wavelength and a **spectrum** is generated by plotting absorbance as a function of wavelength. Absorption of UV light causes an **electron transition**, or excitation, from the ground state to a higher energy level. In the ground state, π **electrons** from double bonds are in the π **bonding** molecular orbital, and **nonbonding electrons** (ie, lone electron pairs) are in the **n nonbonding** molecular orbital. The nonbonding molecular orbitals are higher in energy than the π bonding molecular orbitals.

Upon interactions with UV light of sufficient energy, π and nonbonding electrons are excited to the Lowest Unoccupied Molecular Orbital (LUMO), called the π^* **antibonding** orbital. These excitations are described as a transition from the lower energy bonding molecular orbital to the higher energy antibonding orbital in the form of $\pi \rightarrow \pi^*$ or $n \rightarrow \pi^*$, depending on whether a π or n electron is excited.

The lone electron pairs on nitrogen in pyrimidine are nonbonding (n) electrons. When absorbing UV light, these electrons are excited to the antibonding (π^*) orbital and therefore undergo a **$n \rightarrow \pi^*$** transition.

(Choice A) A $\pi \rightarrow \pi^*$ electron transition describes excitation of a π electron from the bonding to the antibonding molecular orbital rather than a nonbonding electron to an antibonding molecular orbital.

(Choices C and D) The $\pi^* \rightarrow \pi$ and $\pi^* \rightarrow n$ electron transitions describe an excited π electron and excited nonbonding electron, respectively, falling from the antibonding orbital to its ground state (π and n).

Educational objective:

Ultraviolet (UV) light is electromagnetic radiation that corresponds to wavelengths between 200 nm and 400 nm, and UV absorption causes an electron transition from the ground state to a higher energy level. In the ground state, π electrons from double bonds are in the π bonding molecular orbital, and nonbonding electrons are in the n nonbonding molecular orbital. These electrons can be excited to the π^* antibonding orbital via $\pi \rightarrow \pi^*$ or $n \rightarrow \pi^*$ transitions.

Time spent: 0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical Methods